SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Code of Conduct

I S. Dubasri (192511296 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: d Dubasri

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

17) Compare Type-1 and Type-2 Hypervisors for a cloud Data Center Hosting Mission-critical Applications.

Features	Type-1 Hypervisor	Type-2 Hypervisor.
* Installation	Runs directly on hardware	Runs on top of a host operating System
* performance	very high	Lower due to OS overhead.
* Security	More Secure with Smaller attack surface	Less secure because host OS can be attacked
* Manageability	Better for large cloud environments	suitable for personal or Testing use.

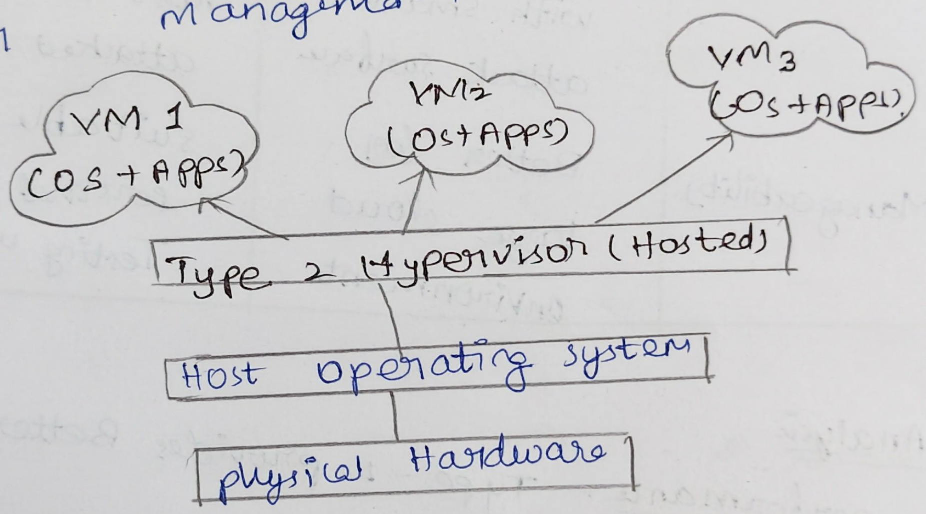
Analysis

* performance: Type-1 provides Better CPU, memory, and I/O performance

* Security: Type-1 is more secure since there is no host OS Layer.

* **Manageability:**
supports centralized Management, Live Migration and high availability.

Justification:-
Type -1 hypervisor is the best choice for Mission - critical cloud applications because it offers higher performance, stronger security and Easier management.



conclusion
for Mission critical cloud application, Type -1 Hypervisor is the best choice because it provides maximum performance ~

2) virtualization Architecture for workload isolation.

cloud provider hosts: * Finance Application, * Healthcare Applications * student portal.

all run on the same physical server.

How virtualization provides Isolation.

* Every application runs inside it own virtual Machine.
* Each VM has its own operating system.
* CPU, RAM, storage, and Network resources are Isolated.
* If one VM crashes, the other continue running safely.

Benefits
* Better security
* Efficient hardware utilization.
* Independent resource allocation.
* Easier Backup and recovery

Major attack surface

1. Hyperviser Attack
* Attackers exploit hyperviser vulnerabilities
* can affect Multiple virtual Machines.

2. VM Escape Attack

* Malware escapes from one VM to access the host or other VMs.

Virtualization enables secure multi-tenant cloud environments while maintaining strong workload isolation.

3. VM Host Utilization Analysis

Given values

* CPU utilization = 92%
* Memory utilization = 88%
* Storage I/O Latency = 35ms
* VM Boot Time = 170 second.

Analysis :-

CPU (92%) → CPU is heavily loaded and Applications become slow.
→ New VMs may not get enough processing power.

Memory (88%).
→ RAM is almost full.
→ System may start swapping memory to disk.
→ performance decrease.

4) ## software Defined Data center (SDDC)

A software Defined Datacenter (SDDC) manages compute, storage and networking entirely Through software.

Traditional Data center.

* Hardware configured Manually.
* Slow Deployment.
* Limited automation.
* Difficult to scale.

Software Defined Data center.

* software controls all resources.
* Automated provisioning.
* Fast deployment.
* Easy scalability.
* centralized Management.

virtualization components.

compute virtualization.

* create Multiple VMs on one server
* Improves CPU utilization.

storage virtualization.

* creates multiple VMs on one server pool.
* Many storage devices into one storage pool.
* Flexible storage allocation.

Storage Latency (35MS).

* storage responds slowly
* File access becomes delayed
* Database performance is affected

VM Boot Time (170s)

* VM startup is very slow.
* Maily caused by storage and CPU Bottlenecks.

Corrective Actions.

* Add more CPU cores.
* Increase RAM.
* Upgrade HDD to SSD / NVMe.
* Balance VMs across multiple hosts.
* Enable resource monitoring and auto-scaling

Likely Bottlenecks.

* CPU overLoad
* Memory shortage
* slow storage subsystem.

Network virtualization

* Creates virtual switches and virtual Network.
* Managed using software Defined Networking (SDN).

Advantages.

Faster Deployment.
Lower operational cost.
Better resource utilization.
High Availability
Easier Disaster recovery

5. Multi-cloud Identity Federation.

Problem:-

A company uses multiple cloud provider. Each provider has its own identity system, causing:
* Login failures.
* Duplicate user accounts.
* permission Mismatches.
* security issues.

proposed secure Design.

Identity Federation.

* shares user Identity across different cloud providers

Single Sign-On (SSO).

* User log in once and access all cloud services.

Authentication protocols.

* SAML, * OAuth 2.0, * OpenID Connect (OIDC)

Security Measures.

* Multi-factor Authentication (MFA).
* Encryption of identity information.
* Role-Based Access control (RBAC)
* Regular audit logs.
* Least privilege access.

Benefits.

* Secure authentication.
* Better privacy protection
* Reduced password management.
* Easy access across multiple cloud platforms.

Conclusion:-

Identity federation using SSO, SAML/OAuth/OIDC, MFA, and RBAC provides secure, seamless access across Multiple cloud provider while protecting user privacy